

P8109 Statistical Inference – Homework 1

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2026-02-04

Problem 1

Let Z_i , $i = 1, \dots, 6$, be iid $N(0, 1)$ random variables. Using a statistical software, calculate

$$P\left(\sum_{i=1}^6 Z_i^2 < 6\right), \quad P\left(\sum_{i=1}^4 Z_i^2 > 3 \sum_{i=5}^6 Z_i^2\right).$$

Solution.

Since $Z_i \stackrel{\text{iid}}{\sim} N(0, 1)$, we have $\sum_{i=1}^6 Z_i^2 \sim \chi_6^2$. Thus

$$P\left(\sum_{i=1}^6 Z_i^2 < 6\right) = F_{\chi_6^2}(6).$$

For the second probability, let $U = \sum_{i=1}^4 Z_i^2 \sim \chi_4^2$ and $V = \sum_{i=5}^6 Z_i^2 \sim \chi_2^2$, which are independent. Then

$$P(U > 3V) = P\left(\frac{U}{V} > 3\right) = P\left(\frac{U/4}{V/2} > \frac{3}{2}\right) = P\left(F > \frac{3}{2}\right),$$

where $F = \frac{U/4}{V/2} \sim F_{4,2}$.

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# First probability
pchisq(6, df = 6)
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## [1] 0.5768099
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# Second probability
1 - pf(3/2, df1 = 4, df2 = 2)
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## [1] 0.4375
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Problem 2

Let $X \sim F_{\nu_1, \nu_2}$. **Show that** $Y = 1/X \sim F_{\nu_2, \nu_1}$.

Proof. By definition, $X \sim F_{\nu_1, \nu_2}$ means

$$X = \frac{U/\nu_1}{V/\nu_2},$$

where $U \sim \chi_{\nu_1}^2$, $V \sim \chi_{\nu_2}^2$, and $U \perp V$. Then

$$Y = \frac{1}{X} = \frac{V/\nu_2}{U/\nu_1}.$$

Since $V \sim \chi_{\nu_2}^2$ and $U \sim \chi_{\nu_1}^2$ are independent, this is exactly the definition of F_{ν_2, ν_1} .

Problem 3

Let $T = Z/\sqrt{W/\nu}$, where $Z \sim N(0, 1)$ and $W \sim \chi_\nu^2$ are independent.

(a) Write down the distribution of T .

By definition (Eq. 1.4), $T \sim t_\nu$ (the Student's t -distribution with ν degrees of freedom).

(b) Show $E T = 0$ for $\nu > 1$ and $\text{Var}(T) = \nu/(\nu - 2)$ for $\nu > 2$.

Given: $E(W^\alpha) = \frac{2^\alpha \Gamma(\nu/2 + \alpha)}{\Gamma(\nu/2)}$, valid for $\nu > -2\alpha$.

i. Since Z and W are independent,

$$E T = E \left[\frac{Z}{\sqrt{W/\nu}} \right] = E[Z] \cdot E \left[\frac{\sqrt{\nu}}{\sqrt{W}} \right] = 0 \cdot E[W^{-1/2}] \cdot \sqrt{\nu} = 0,$$

provided $E[W^{-1/2}] < \infty$. Using the given formula with $\alpha = -1/2$:

$$E[W^{-1/2}] = \frac{2^{-1/2} \Gamma(\nu/2 - 1/2)}{\Gamma(\nu/2)},$$

which is finite when $\nu/2 - 1/2 > 0$, i.e., $\nu > 1$.

ii. Since $E T = 0$ for $\nu > 1$,

$$\text{Var}(T) = E[T^2] = E \left[\frac{Z^2 \nu}{W} \right] = \nu E[Z^2] E[W^{-1}] = \nu E[W^{-1}],$$

using independence and $E[Z^2] = 1$. Applying the given formula with $\alpha = -1$:

$$E[W^{-1}] = \frac{2^{-1} \Gamma(\nu/2 - 1)}{\Gamma(\nu/2)} = \frac{1}{2} \cdot \frac{1}{\nu/2 - 1} = \frac{1}{\nu - 2},$$

where we used $\Gamma(\nu/2) = (\nu/2 - 1) \Gamma(\nu/2 - 1)$. This requires $\nu/2 - 1 > 0$, i.e., $\nu > 2$.

Therefore $\text{Var}(T) = \nu/(\nu - 2)$ for $\nu > 2$.

Problem 4

CLT for the Poisson distribution: if $X \sim \text{Poisson}(\lambda)$, then $(X - \lambda)/\sqrt{\lambda} \xrightarrow{d} N(0, 1)$ as $\lambda \rightarrow \infty$.

(a) Show $E e^{tX^*} = e^{-t\sqrt{\lambda} - \lambda + \lambda e^{t/\sqrt{\lambda}}}$, **where** $X^* = (X - \lambda)/\sqrt{\lambda}$.

Proof. The MGF of $X \sim \text{Poisson}(\lambda)$ is $M_X(s) = E[e^{sX}] = e^{\lambda(e^s - 1)}$. Now

$$E[e^{tX^*}] = E\left[\exp\left(\frac{t(X - \lambda)}{\sqrt{\lambda}}\right)\right] = e^{-t\sqrt{\lambda}} E\left[\exp\left(\frac{t}{\sqrt{\lambda}} X\right)\right] = e^{-t\sqrt{\lambda}} M_X\left(\frac{t}{\sqrt{\lambda}}\right).$$

Substituting the Poisson MGF:

$$= e^{-t\sqrt{\lambda}} \exp\left[\lambda(e^{t/\sqrt{\lambda}} - 1)\right] = \exp(-t\sqrt{\lambda} - \lambda + \lambda e^{t/\sqrt{\lambda}}).$$

(b) Show $E e^{tX^*} \rightarrow e^{t^2/2}$ **as** $\lambda \rightarrow \infty$.

Proof. Let $u = t/\sqrt{\lambda} \rightarrow 0$ as $\lambda \rightarrow \infty$. Taylor-expand e^u :

$$e^u = 1 + u + \frac{u^2}{2} + \frac{u^3}{6} + \dots$$

Multiply by λ :

$$\lambda e^{t/\sqrt{\lambda}} = \lambda + t\sqrt{\lambda} + \frac{t^2}{2} + \frac{t^3}{6\sqrt{\lambda}} + \dots$$

Therefore

$$-t\sqrt{\lambda} - \lambda + \lambda e^{t/\sqrt{\lambda}} = \frac{t^2}{2} + \frac{t^3}{6\sqrt{\lambda}} + \dots \rightarrow \frac{t^2}{2} \text{ as } \lambda \rightarrow \infty.$$

Hence $E[e^{tX^*}] \rightarrow e^{t^2/2}$, which is the MGF of $N(0, 1)$. By Levy's continuity theorem (Theorem 2.2.4), $X^* \xrightarrow{d} N(0, 1)$.

Problem 5

Prove from first principles that if $X \sim \text{Binomial}(n, p)$, then $\frac{X - np}{\sqrt{np(1-p)}} \xrightarrow{d} N(0, 1)$.

Proof. Write $X =_d Y_1 + \dots + Y_n$ where $Y_i \stackrel{\text{iid}}{\sim} \text{Bernoulli}(p)$, with $E[Y_i] = p$ and $\text{Var}(Y_i) = p(1-p) =: \sigma^2$.

Define $Z_n = \frac{X - np}{\sigma\sqrt{n}} = \frac{\sum_{i=1}^n (Y_i - p)}{\sigma\sqrt{n}}$. The MGF of Z_n is

$$M_{Z_n}(t) = \left[E \exp\left(\frac{t(Y_1 - p)}{\sigma\sqrt{n}}\right) \right]^n$$

by independence. Let $s = t/(\sigma\sqrt{n})$. Then

$$\begin{aligned} E[e^{s(Y_1 - p)}] &= p e^{s(1-p)} + (1-p) e^{-sp} \\ &= p \left(1 + s(1-p) + \frac{s^2(1-p)^2}{2} + O(s^3) \right) \\ &\quad + (1-p) \left(1 - sp + \frac{s^2 p^2}{2} + O(s^3) \right) \\ &= 1 + s \underbrace{[p(1-p) - (1-p)p]}_{=0} + \frac{s^2}{2} \underbrace{[p(1-p)^2 + (1-p)p^2]}_{=p(1-p)=\sigma^2} + O(s^3) \\ &= 1 + \frac{s^2 \sigma^2}{2} + O(s^3). \end{aligned}$$

Substituting $s^2 \sigma^2 = t^2/n$:

$$E[e^{s(Y_1 - p)}] = 1 + \frac{t^2}{2n} + o\left(\frac{1}{n}\right).$$

Therefore

$$M_{Z_n}(t) = \left[1 + \frac{t^2}{2n} + o\left(\frac{1}{n}\right) \right]^n \longrightarrow e^{t^2/2} \quad \text{as } n \rightarrow \infty,$$

using the standard limit $(1 + a_n/n)^n \rightarrow e^{a_n}$ when $a_n \rightarrow a$. Since $e^{t^2/2}$ is the MGF of $N(0, 1)$, by L'evy's continuity theorem, $Z_n \xrightarrow{d} N(0, 1)$.

Problem 6

Let $Y_1, \dots, Y_5 \stackrel{\text{iid}}{\sim} N(0, 1)$ with sample mean $\bar{Y}_5$. Let Y_6 be an independent $N(0, 1)$ variable.

(a) $\bar{Y}_5$

$\bar{Y}_5 = \frac{1}{5} \sum_{i=1}^5 Y_i$. By Theorem 1.2, a linear combination of independent normals is normal:

$$E[\bar{Y}_5] = 0, \quad \text{Var}(\bar{Y}_5) = \frac{1}{25} \sum_{i=1}^5 1 = \frac{1}{5}.$$

Therefore $\boxed{\bar{Y}_5 \sim N(0, 1/5)}$.

(b) $U = \sum_{i=1}^5 Y_i^2$

Each $Y_i \sim N(0, 1)$, so $Y_i^2 \sim \chi_1^2$. Since they are independent,

$$\boxed{U = \sum_{i=1}^5 Y_i^2 \sim \chi_5^2.}$$

(c) $V = \sum_{i=1}^5 (Y_i - \bar{Y}_5)^2$

Note $V = (n-1)S^2$ with $n = 5$ and $\sigma^2 = 1$. By Theorem 1.4(c),

$$(n-1)S^2/\sigma^2 = V \sim \chi_{n-1}^2.$$

Therefore $\boxed{V \sim \chi_4^2}$.

(d) $W = \sum_{i=1}^5 (Y_i - \bar{Y}_5)^2 + Y_6^2$

We have $V \sim \chi_4^2$ (from part (c)) and $Y_6^2 \sim \chi_1^2$. Since V depends only on $Y_1, \dots, Y_5$ and Y_6 is independent of them, V and Y_6^2 are independent. By the additivity of independent chi-squared random variables,

$$\boxed{W = V + Y_6^2 \sim \chi_5^2.}$$

Problem 7

The coefficient of variation is $CV = \sigma/\mu$. The sample coefficient of variation is $\widehat{CV} = S/\bar{Y}$ where $Y_i \sim N(\mu, \sigma^2)$.

(a) Find the distribution of $\widehat{CV}$.

By Theorem 1.4, $\bar{Y}$ and S^2 are independent with

$$\bar{Y} \sim N\left(\mu, \frac{\sigma^2}{n}\right), \quad \frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2.$$

Define $Z = \frac{\bar{Y}}{\sigma/\sqrt{n}} \sim N\left(\frac{\sqrt{n}\mu}{\sigma}, 1\right)$ and $V = \frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$, which are independent. Then

$$\frac{\sqrt{n}}{\widehat{CV}} = \frac{\sqrt{n}\bar{Y}}{S} = \frac{\bar{Y}/(\sigma/\sqrt{n})}{S/\sigma} = \frac{Z}{\sqrt{V/(n-1)}},$$

which by definition follows a **noncentral t -distribution** with $n-1$ degrees of freedom and noncentrality parameter

$$\delta = \frac{\sqrt{n}\mu}{\sigma} = \frac{\sqrt{n}}{CV}.$$

Therefore:

$$\boxed{\frac{\sqrt{n}}{\widehat{CV}} \sim t_{n-1}\left(\frac{\sqrt{n}}{CV}\right)},$$

i.e., $\widehat{CV} = \sqrt{n}/T$ where $T \sim t_{n-1}(\sqrt{n}/CV)$. In other words, $\widehat{CV}$ follows the distribution of a scaled reciprocal of a noncentral t -distributed random variable.

(b) Find c such that $P(|\widehat{CV}| \leq c) = 0.95$.

Since $\widehat{CV} = \sqrt{n}/T$,

$$P(|\widehat{CV}| \leq c) = P\left(|T| \geq \frac{\sqrt{n}}{c}\right) = 0.95.$$

Equivalently, $P(|T| < \sqrt{n}/c) = 0.05$, where $T \sim t_{n-1}(\sqrt{n}/CV)$. Denoting the 2.5th percentile of this noncentral t by $t_{0.025}$ and the 97.5th percentile by $t_{0.975}$,

$$c = \frac{\sqrt{n}}{\min(|t_{0.025}|, t_{0.975})}.$$

Problem 8

Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$ and $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$.

(a) Identify the distribution of S^2 .

By Theorem 1.4(c), $\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$. Therefore

$$S^2 \sim \frac{\sigma^2}{n-1} \chi_{n-1}^2,$$

i.e., S^2 is a scaled chi-squared. Equivalently, $S^2 \sim \text{Gamma}\left(\frac{n-1}{2}, \frac{2\sigma^2}{n-1}\right)$.

(b) Show $\sqrt{n} \left(\frac{S^2}{\sigma^2} - 1 \right) \xrightarrow{d} N(0, 2)$.

Proof. Let $V = \frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$. Write $V = \sum_{i=1}^{n-1} W_i$ where $W_i \stackrel{\text{iid}}{\sim} \chi_1^2$ with $E[W_i] = 1$ and $\text{Var}(W_i) = 2$.

By the Lindeberg-L'evy CLT (Theorem 1.3),

$$\frac{V - (n-1)}{\sqrt{2(n-1)}} \xrightarrow{d} N(0, 1).$$

Now

$$\sqrt{n} \left(\frac{S^2}{\sigma^2} - 1 \right) = \sqrt{n} \cdot \frac{V - (n-1)}{n-1} = \sqrt{\frac{2n}{n-1}} \cdot \frac{V - (n-1)}{\sqrt{2(n-1)}}.$$

Since $\sqrt{2n/(n-1)} \rightarrow \sqrt{2}$ as $n \rightarrow \infty$ and $\frac{V - (n-1)}{\sqrt{2(n-1)}} \xrightarrow{d} N(0, 1)$, by Slutsky's theorem (Theorem 2.5):

$$\sqrt{n} \left(\frac{S^2}{\sigma^2} - 1 \right) \xrightarrow{d} \sqrt{2} \cdot N(0, 1) = N(0, 2).$$

Problem 9

Verify whether each distribution belongs to the exponential family.

(a) Binomial(n, p)

The pmf is (with n fixed and p as the parameter):

$$f(x | p) = \binom{n}{x} p^x (1-p)^{n-x} = \binom{n}{x} \exp \left\{ x \log \frac{p}{1-p} + n \log(1-p) \right\}.$$

This has the form $h(x) \exp\{c(p) T(x) - A(p)\}$ with:

$$h(x) = \binom{n}{x}, \quad T(x) = x, \quad c(p) = \log \frac{p}{1-p}, \quad A(p) = -n \log(1-p).$$

Since $c(p)$ is strictly monotone in $p \in (0, 1)$, this is a **one-parameter exponential family** with natural parameter $\eta = \log \frac{p}{1-p}$.

(b) Gamma(α, β)

The pdf is ($\alpha, \beta > 0$):

$$f(x | \alpha, \beta) = \frac{1}{\Gamma(\alpha) \beta^\alpha} x^{\alpha-1} e^{-x/\beta}, \quad x > 0.$$

Rewrite as:

$$f(x | \alpha, \beta) = \frac{1}{x} \mathbb{I}(x > 0) \cdot \exp \left\{ (\alpha - 1) \log x + \left(-\frac{1}{\beta} \right) x - \log \Gamma(\alpha) - \alpha \log \beta \right\}.$$

This has the form $h(x) \exp\{\sum_{j=1}^2 c_j(\theta) T_j(x) - A(\theta)\}$ with:

$$h(x) = \frac{\mathbb{I}(x > 0)}{x}, \quad \mathbf{T}(x) = \begin{pmatrix} \log x \\ x \end{pmatrix}, \quad \mathbf{c}(\theta) = \begin{pmatrix} \alpha - 1 \\ -1/\beta \end{pmatrix}, \quad A(\theta) = \log \Gamma(\alpha) + \alpha \log \beta.$$

Since $\mathbf{c}(\alpha, \beta)$ maps $(\alpha, \beta) \in (0, \infty)^2$ onto an open subset of $\mathbb{R}^2$, this is a **full-rank two-parameter exponential family**.

Problem 10

(a) Find $h(X)$ such that $h(\mu) + \frac{1}{2}h''(\mu)\text{Var}(X)$ is not a good approximation of $\mathbb{E}[h(X)]$.

Example. Let $h(x) = e^x$ and $X \sim N(\mu, \sigma^2)$.

- **Exact:** $\mathbb{E}[e^X] = e^{\mu + \sigma^2/2}$ (by the normal MGF).
- **Approximation:** $h(\mu) + \frac{1}{2}h''(\mu)\sigma^2 = e^\mu + \frac{1}{2}e^\mu\sigma^2 = e^\mu(1 + \frac{\sigma^2}{2})$.

The ratio of exact to approximate is $\frac{e^{\sigma^2/2}}{1 + \sigma^2/2}$, which diverges as σ^2 grows. For instance, with $\mu = 0$ and $\sigma^2 = 4$:

- Exact: $e^{4/2} = e^2 \approx 7.389$.
- Approximation: $1 + 4/2 = 3$.

The approximation fails because $h(x) = e^x$ has rapidly growing higher-order derivatives ($h^{(k)}(x) = e^x$ for all k), making the neglected higher-order terms in the Taylor expansion significant when σ^2 is not small.

(b) Prove the expansion $\mathbb{E}[h(\bar{X})] = h(\mu) + \frac{\sigma^2}{2n}h''(\mu) + R_n$ where $R_n = O(1/n^2)$.

Proof. Let $X_1, \dots, X_n$ be i.i.d. with mean μ , variance σ^2 , and finite fourth moment $\mu_4 = \mathbb{E}[(X_1 - \mu)^4] < \infty$. By Taylor's theorem with Lagrange remainder, for some ξ between $\bar{X}$ and μ :

$$h(\bar{X}) = h(\mu) + h'(\mu)(\bar{X} - \mu) + \frac{h''(\mu)}{2}(\bar{X} - \mu)^2 + \frac{h'''(\mu)}{6}(\bar{X} - \mu)^3 + \frac{h^{(4)}(\xi)}{24}(\bar{X} - \mu)^4.$$

Taking expectations and using $\mathbb{E}[\bar{X} - \mu] = 0$ and $\mathbb{E}[(\bar{X} - \mu)^2] = \sigma^2/n$:

$$\mathbb{E}[h(\bar{X})] = h(\mu) + \frac{\sigma^2}{2n}h''(\mu) + \underbrace{\frac{h'''(\mu)}{6}\mathbb{E}[(\bar{X} - \mu)^3] + \frac{1}{24}\mathbb{E}[h^{(4)}(\xi)(\bar{X} - \mu)^4]}_{=: R_n}.$$

We show $R_n = O(1/n^2)$:

1. **Third-moment term.** By independence and $\mathbb{E}[X_i - \mu] = 0$:

$$\mathbb{E}[(\bar{X} - \mu)^3] = \frac{1}{n^3} \mathbb{E} \left[\left(\sum_{i=1}^n (X_i - \mu) \right)^3 \right] = \frac{1}{n^3} \sum_{i=1}^n \mathbb{E}[(X_i - \mu)^3] = \frac{\mu_3}{n^2},$$

where $\mu_3 = \mathbb{E}[(X_1 - \mu)^3]$ is finite (since $|\mu_3| \leq \mu_4^{3/4}$ by Hölder's inequality, or directly by the existence of the fourth moment). The cross terms vanish by independence and zero mean. Hence this term is $O(1/n^2)$.

2. **Fourth-moment term.** Since $|h^{(4)}(\xi)| \leq M$,

$$\left| \frac{1}{24} \mathbb{E}[h^{(4)}(\xi)(\bar{X} - \mu)^4] \right| \leq \frac{M}{24} \mathbb{E}[(\bar{X} - \mu)^4].$$

Expanding $(\sum(X_i - \mu))^4$ and using independence:

$$\mathbb{E} \left[\left(\sum_{i=1}^n (X_i - \mu) \right)^4 \right] = n\mu_4 + 3n(n-1)\sigma^4.$$

Therefore

$$\mathbb{E}[(\bar{X} - \mu)^4] = \frac{n\mu_4 + 3n(n-1)\sigma^4}{n^4} = \frac{\mu_4}{n^3} + \frac{3(n-1)\sigma^4}{n^3} = O(1/n^2).$$

Combining, $R_n = O(1/n^2)$.